

RAROTONGA, Cook Islands

WMO: 918430

Lat: 21.2009S

Lon: 159.8130W

Elev: 19

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	62.9	64.7	51.7	57.0	69.0	53.5	60.9	69.1	25.0	72.8	22.3	73.4	3.8	160	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.7	87.5	80.2	86.1	79.2	85.2	78.5	81.3	85.3	80.3	84.2	79.7	83.5	8.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	157.1	84.9	79.1	151.0	83.5	78.6	148.5	82.9	45.1	85.4	44.0	85.0	43.3	83.3	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.9	17.7	59.5	90.2	1.7	1.9	58.2	91.6	57.2	92.7	56.2	93.7	55.0	95.1		
(5)			55.6	84.3	1.6	2.1	54.4	85.8	53.5	87.0	52.5	88.2	51.4	89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.4	79.6	80.3	80.3	78.7	76.1	74.2	72.9	72.5	73.4	74.6	76.9	78.2	(6)
(7)		DBStd	3.76	2.00	2.02	2.02	2.30	2.68	2.86	2.90	2.92	2.91	2.60	2.46	2.24	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	9653	919	847	939	860	808	725	709	698	702	764	807	875	(10)
(11)		CDD65	4178	454	427	474	410	343	275	244	233	252	299	357	410	(11)
(12)		CDH74	28281	4158	4121	4496	3282	1971	1073	769	686	859	1324	2350	3192	(12)
(13)		CDH80	4556	835	950	1025	565	170	61	24	20	39	78	287	502	(13)
(14)	Wind	WSAvg	8.7	8.8	8.9	8.2	8.5	8.3	8.4	8.7	8.8	9.4	8.9	8.6	9.2	(14)
(15)	Precipitation	PrecAvg	85.7	10.3	9.0	8.4	9.0	7.7	4.6	4.0	5.4	5.0	4.5	6.3	8.9	(15)
(16)		PrecMax	154.5	43.3	24.8	18.7	22.3	27.3	12.0	9.8	18.7	14.3	13.1	18.0	26.5	(16)
(17)		PrecMin	46.7	0.9	0.5	0.5	1.2	0.7	0.3	0.9	0.7	0.7	0.4	0.8	0.2	(17)
(18)		PrecStd	24.1	7.3	4.6	4.9	4.4	5.5	3.2	2.2	3.7	3.4	3.0	4.6	6.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.8	89.4	89.2	87.6	85.8	84.0	82.5	82.2	82.8	84.1	86.2	87.5	(19)
MCWB			80.6	81.8	81.3	80.6	79.3	77.9	76.3	75.5	76.2	77.1	78.6	79.7	(20)	
2%		DB	86.1	86.8	86.8	85.8	82.8	81.6	80.4	80.3	80.9	82.1	84.2	84.9	(21)	
		MCWB	79.1	79.5	79.9	79.0	76.7	75.8	74.6	74.3	75.0	75.6	76.9	77.9	(22)	
5%		DB	84.6	85.7	85.8	84.4	82.0	80.3	79.0	78.8	79.2	80.6	82.7	84.1	(23)	
		MCWB	78.0	78.8	79.2	77.9	75.9	74.7	73.3	73.1	73.4	74.3	75.9	77.1	(24)	
10%		DB	84.1	84.4	84.4	82.9	80.6	78.8	77.5	77.3	78.4	79.1	81.8	82.6	(25)	
		MCWB	77.6	78.1	78.2	76.9	74.8	73.6	72.3	72.0	72.6	72.9	75.3	76.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	83.1	82.6	81.7	80.1	78.9	77.7	77.3	78.1	79.0	79.9	81.2	(27)
MCD B			85.7	87.3	87.1	85.6	84.1	82.4	80.7	80.4	81.2	82.1	83.8	85.1	(28)	
2%		WB	80.3	80.9	81.0	80.0	78.5	77.2	76.2	75.8	76.3	77.3	78.5	79.5	(29)	
		MCD B	84.1	84.9	85.1	83.7	81.7	80.4	79.0	78.7	79.5	80.4	82.4	83.2	(30)	
5%		WB	79.4	79.9	80.0	79.2	77.4	76.0	74.9	74.6	75.1	76.0	77.5	78.6	(31)	
		MCD B	83.3	83.8	84.0	82.9	80.4	79.1	77.8	77.7	78.4	79.3	81.3	82.3	(32)	
10%		WB	78.7	79.1	79.2	78.3	76.3	74.6	73.8	73.4	73.9	74.7	76.5	77.7	(33)	
		MCD B	82.7	83.0	83.2	82.1	79.5	78.0	77.0	76.8	77.5	78.1	80.3	81.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	8.7	8.5	8.7	8.5	8.7	9.0	8.6	8.9	9.0	8.5	(35)
MCDBR			9.2	9.8	10.0	9.9	9.8	9.6	9.7	9.7	9.1	9.5	9.8	9.8	(36)	
MCWBR			5.5	5.7	5.8	6.0	6.1	6.1	6.6	6.6	6.0	5.9	5.8	5.6	(37)	
5% WB		MCDBR	8.5	8.8	9.2	9.3	8.6	8.5	8.4	8.8	8.5	8.5	8.8	8.7	(38)	
		MCWBR	5.6	5.9	6.0	6.2	6.3	6.4	7.0	7.0	6.5	6.2	5.9	5.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.385	0.384	0.375	0.379	0.373	0.365	0.362	0.361	0.381	0.375	0.377	0.382	(40)	
taud		2.612	2.616	2.650	2.633	2.626	2.628	2.625	2.622	2.563	2.577	2.598	2.598	(41)		
Ebn at Noon		305	301	296	282	271	267	272	284	289	300	305	305	(42)		
Edh at Noon		33	32	30	28	27	25	26	28	32	33	33	33	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2012	1947	1760	1495	1218	1134	1200	1423	1688	1899	2014	2025	(44)	
(45)		RadStd	162	152	128	96	107	88	80	79	105	146	160	166	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.45	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+223	+221
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page